MIL-S-3786/38A(USAF)
15 OCTOBER 1980
SUPERSEDING
MIL-S-3786/38(USAF)
23 October 1973

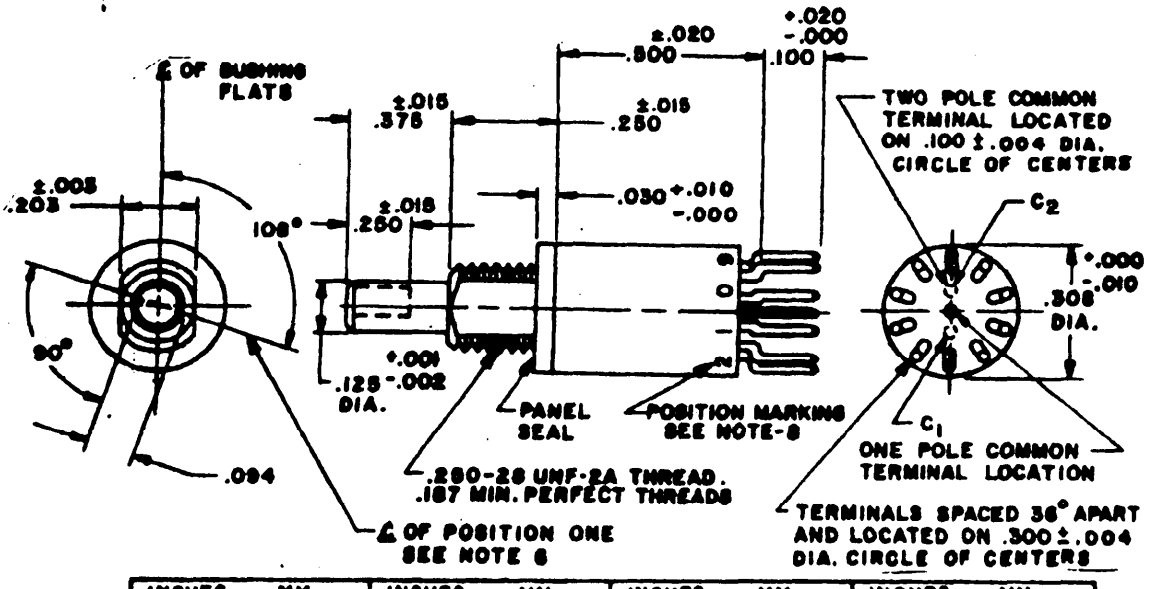
# MILITARY SPECIFICATION SHEET

## SWITCH, ROTARY, CLOSED CONSTRUCTION, 100 MILLIAMPERES

This specification (sheet) is approved for use by the Air Force Acquisition Logistics Division, Electronic Support Division (AFALD/PTES), Wright-Patterson AFB, Ohio 45433, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of specification MIL-S-3786.

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION. USE MIL-S-3786/20.

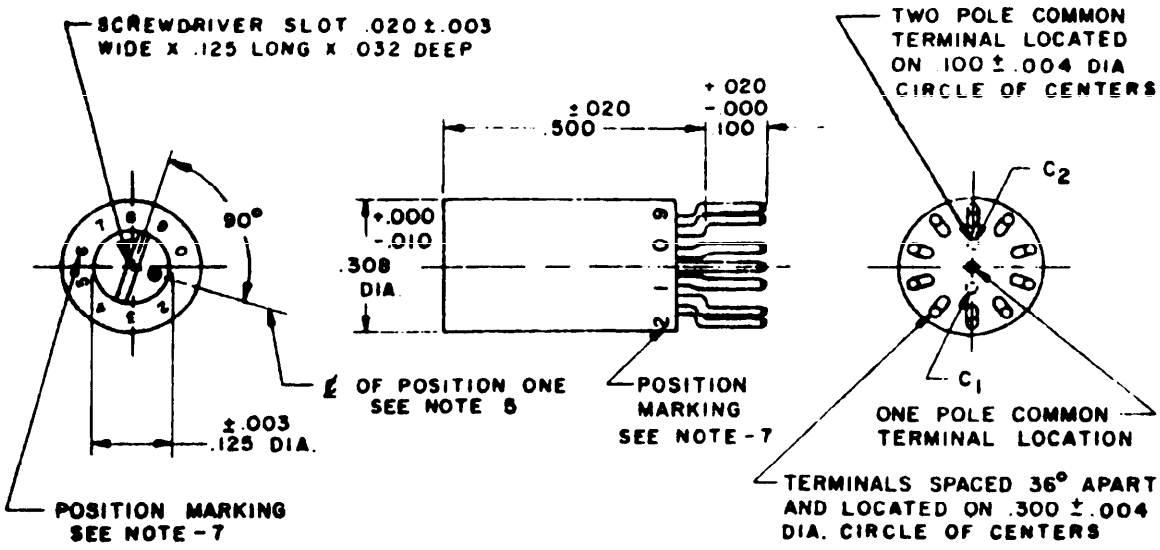



| INCHES | MM | INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|---|---|
| .001 | .03 | | | | | | |
| .002 | .05 | .010 | .25 | .094 | 2.39 | .250 | 6.35 |
| .003 | .08 | .015 | .38 | .100 | 2.54 | .300 | 7.62 |
| .004 | .10 | .020 | .51 | .125 | 3.18 | .308 | 7.82 |
| .005 | .13 | .030 | .76 | .203 | 5.16 | .500 | 12.70 |

NOTES:
1) DIMENSIONS ARE IN INCHES.
2) METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND BASED UPON 1 INCH = 25.4 MM.
3) UNLESS OTHERWISE SPECIFIED TOLERANCES ARE ±.005 (.13 MM) AND ±3° ON ANGLES (NON-ACCUMULATIVE).
4) TERMINAL DIAMETER .015 ±.002
5) MOUNTING HARDWARE: EACH SWITCH SHALL BE PROVIDED WITH ONE HEXAGON NUT, UNLESS OTHERWISE SPECIFIED, .312 ±.010 INCH ACROSS FLATS AND .062 ±.010 INCH THICK; ONE INTERNAL TOOTH LOCKWASHER .415 MAX. O.D.
6) SHAFT FLAT IS LOCATED DIAMETRICALLY OPPOSITE TO THE NUMBER ONE TERMINAL WHEN SWITCH IS IN POSITION ONE.
7) POSITION ONE COINCIDES WITH TERMINAL ONE.
8) COLOR OF SWITCH BODY SHALL BE BLACK WITH WHITE POSITION MARKING ORIENTED AS SHOWN.

FIGURE 1. Shaft actuated.

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| INCHES | MM | INCHES | MM | INCHES | MM |
|---|---|---|---|---|---|
| .002 | .05 | | | | |
| .003 | .08 | .015 | .38 | .125 | 3.18 |
| .004 | .10 | .020 | .51 | .300 | 7.62 |
| .005 | .13 | .032 | .81 | .308 | 7.82 |
| .010 | .25 | .100 | 2.54 | .500 | 12.70 |

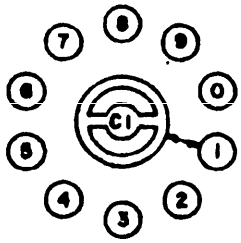
NOTES:

1) DIMENSIONS ARE IN INCHES.
2) METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
3) UNLESS OTHERWISE SPECIFIED TOLARANCES ARE ±.005 (.13 MM) AND ±3° ON ANGLES (NON-ACCUMULATIVE).
4) TERMINAL DIAMETER .015 ±.002 .
5) POSITION LOCATOR MARK APPEARS ADJACENT TO THE NUMBER ONE TERMINAL WHEN SWITCH IS IN POSITION ONE .
6) POSITION ONE COINCIDES WITH TERMINAL ONE.
7) COLOR OF SWITCH BODY SHALL BE BLACK WITH WHITE POSITION MARKING ORIENTED AS SHOWN.

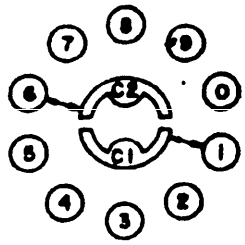
FIGURE 2. Screwdriver actuated.



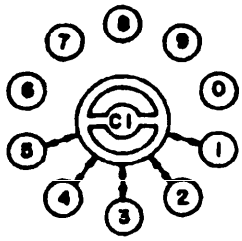
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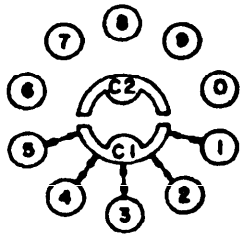

TYPE 1
SINGLE POLE



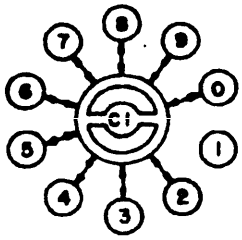

TYPE 2
TWO POLE




TYPE 3
SINGLE POLE




TYPE 4
TWO POLE




TYPE 5
SINGLE POLE

NOTES:
1) CIRCUIT CONFIGURATION VIEWED FROM SHAFT END AND SHOWN IN SWITCH POSITION NUMBER ONE.
2) ALL TERMINALS SHALL BE SUPPLIED REGARDLESS OF NUMBER OF SWITCH POSITIONS.

FIGURE 3. Circuit configuration.



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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, and 3.

Angle of throw: 36°.

Construction: E or F.

Insulation material: Symbol P.

Vibration grade: Symbol 3 (10-2000 Hz).

Shock type: Symbol M (medium impact).

Altitude: Symbol C (70,000 ft.).

Terminal strength: A force of .5 pounds minimum shall be applied to the terminals.

Stop strength: Stops shall be set in a clockwise direction from position number one, when viewed from the shaft end. Stops, when supplied, shall withstand a 2 pound-inch torque.

Rotational torque: The minimum and maximum values of torque determined for shaft rotation shall be within the limits of table I and shall not change more than 50 percent from initial value.

TABLE I. Rotational torque limits.

| Temperature | Torque ounce-inch | |
|---|---|---|
| | Minimum | Maximum |
| Room | 2 | 10 |
| Minimum | 2 | 12 |

Temperature-life characteristic: See tables II and III.

Life (rotational): The test loads for the applicable circuit conditions shall be as specified in tables II and III.

TABLE II. Circuit values for life (rotational) test.
(Applicable to types 1 and 2 of figure 3)

| Environmental conditions | Inductive loads (250 millihenries) | | Resistive loads | |
|---|---|---|---|---|
| | ma | Vdc | ma | Volts |
| At atmospheric pressure and elevated temp. 125°C (25,000 cycles) | 20 | 28 | 50<br>50 | 28 Vdc<br>115 Vrms 60 Hz |
| At reduced barometric pressure (70,000 ft.) and ambient temp. 25°C (25,000 cycles) | 10 | 28 | 25<br>25 | 28 Vdc<br>115 Vrms 60 Hz |
| At atmospheric pressure and ambient temp. 25°C (10,000 cycles) | 30 | 28 | 100<br>75 | 28 Vdc<br>115 Vrms 60 Hz |

Two samples shall be run under each of the loads of table II for qualification. Switch to be tested shall be part number M3786/38-0010.



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Type designation and Style of MIL-S-3786 do not apply due to the reduced number of life (rotational) cycles of table III.

TABLE III. Circuit values for life (rotational) test.
(Applicable to types 3, 4, and 5 of figure 3)

| Environmental conditions | Inductive loads (250 millihenries) | | Resistive loads | |
|---|---|---|---|---|
| | ma | Vdc | ma | Volts |
| At atmospheric pressure and elevated temp. 125°C (10,000 cycles) 1/ | 20 | 28 | 50<br>50 | 28 Vdc<br>115 Vrms 60 Hz |
| At reduced barometric pressure (70,000 ft.) and ambient temp. 25°C (10,000 cycles) 1/ | 10 | 28 | 25<br>25 | 28 Vdc<br>115 Vrms 60 Hz |
| At atmospheric pressure and ambient temp. 25°C (5,000 cycles) 1/ | 30 | 28 | 100<br>75 | 28 Vdc<br>115 Vrms 60 Hz |

1/ Number of life (rotational) cycles derated due to the additional number of rotor contacts per common terminal.

Two samples shall be run under each of the loads of table III for qualification. Each of the loads specified shall be switched by each of the rotor contacts of each switch. Switch to be tested shall be part number M3786/38-0038.

Dielectric withstanding voltage: The applicable voltage specified in table IV shall be applied.

TABLE IV. Dielectric test voltages.

| Altitude | Test voltage (Vrms 60 Hz) |
|---|---|
| At atmospheric pressure | 300 |
| (70,000 ft.) | 150 |

Contact resistance: Initial contact resistance shall not exceed 20 milliohms. After life (rotational) contact resistance shall not exceed 50 milliohms.

Extent of qualification: Figure 1 qualifies figure 2 provided two samples of figure 2 are tested to explosion requirement paragraph to verify explosion proof capability.

Ordering data: Procurement documents should specify the following:

Title, number, and date of this specification sheet, and the military part number as listed in table V.

Part number: See table V.



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TABLE V. Switching characteristic and part number. 3/

| Part no. M3786/38 construction - F Fig. 1 | Part no. M3786/38 construction - E Fig. 2 | Circuit configuration Fig. | No. of positions | 2/ Switching characteristics (S or NS) |
|---|---|---|---|---|
| -0001 | -3001 | 1 | 2 | NS |
| -0002 | -3002 | 1 | 3 | |
| -0003 | -3003 | 1 | 4 | |
| -0004 | -3004 | 1 | 5 | |
| -0005 | -3005 | 1 | 6 | |
| -0006 | -3006 | 1 | 7 | |
| -0007 | -3007 | 1 | 8 | |
| -0008 | -3008 | 1 | 9 | |
| -0009 | -3009 | 1 | 10 1/ | |
| -0010 | -3010 | 1 | 10 | |
| -0011 | -3011 | 1 | 2 | S |
| -0012 | -3012 | 1 | 3 | |
| -0013 | -3013 | 1 | 4 | |
| -0014 | -3014 | 1 | 5 | |
| -0015 | -3015 | 1 | 6 | |
| -0016 | -3016 | 1 | 7 | |
| -0017 | -3017 | 1 | 8 | |
| -0018 | -3018 | 1 | 9 | |
| -0019 | -3019 | 1 | 10 1/ | |
| -0020 | -3020 | 1 | 10 | |
| -0021 | -3021 | 2 | 2 | NS |
| -0022 | -3022 | 2 | 3 | |
| -0023 | -3023 | 2 | 4 | |
| -0024 | -3024 | 2 | 5 | |
| -0025 | -3025 | 2 | 2 | S |
| -0026 | -3026 | 2 | 3 | |
| -0027 | -3027 | 2 | 4 | |
| -0028 | -3028 | 2 | 5 | |
| -0029 | -3029 | 3 | 2 | NS |
| -0030 | -3030 | 3 | 3 | |
| -0031 | -3031 | 3 | 4 | |
| -0032 | -3032 | 3 | 5 | |
| -0033 | -3033 | 3 | 6 | NS |
| -0034 | -3034 | 3 | 7 | |
| -0035 | -3035 | 3 | 8 | |
| -0036 | -3036 | 3 | 9 | |
| -0037 | -3037 | 3 | 10 1/ | |
| -0038 | -3038 | 3 | 10 | |
| -0039 | -3039 | 3 | 2 | S |
| -0040 | -3040 | 3 | 3 | |
| -0041 | -3041 | 3 | 4 | |
| -0042 | -3042 | 3 | 5 | |
| -0043 | -3043 | 3 | 6 | |
| -0044 | -3044 | 3 | 7 | |
| -0045 | -3045 | 3 | 8 | |
| -0046 | -3046 | 3 | 9 | |
| -0047 | -3047 | 3 | 10 1/ | |
| -0048 | -3048 | 3 | 10 | |
| -0049 | -3049 | 4 | 2 | NS |
| -0050 | -3050 | 4 | 3 | |
| -0051 | -3051 | 4 | 4 | |
| -0052 | -3052 | 4 | 5 | |
| -0053 | -3053 | 4 | 6 | |
| -0054 | -3054 | 4 | 7 | |
| -0055 | -3055 | 4 | 8 | |
| -0056 | -3056 | 4 | 9 | |
| -0057 | -3057 | 4 | 10 1/ | |
| -0058 | -3058 | 4 | 10 | |
| -0059 | -3059 | 4 | 2 | S |
| -0060 | -3060 | 4 | 3 | |
| -0061 | -3061 | 4 | 4 | |
| -0062 | -3062 | 4 | 5 | |
| -0063 | -3063 | 4 | 6 | |
| -0064 | -3064 | 4 | 7 | |
| -0065 | -3065 | 4 | 8 | S |
| -0066 | -3066 | 4 | 9 | |
| -0067 | -3067 | 4 | 10 1/ | |
| -0068 | -3068 | 4 | 10 | |
| -0069 | -3069 | 5 | 2 | NS |
| -0070 | -3070 | 5 | 3 | |
| -0071 | -3071 | 5 | 4 | |
| -0072 | -3072 | 5 | 5 | |
| -0073 | -3073 | 5 | 6 | |
| -0074 | -3074 | 5 | 7 | |
| -0075 | -3075 | 5 | 8 | |
| -0076 | -3076 | 5 | 9 | |
| -0077 | -3077 | 5 | 10 1/ | |
| -0078 | -3078 | 5 | 10 | |
| -0079 | -3079 | 5 | 2 | S |
| -0080 | -3080 | 5 | 3 | |
| -0081 | -3081 | 5 | 4 | |
| -0082 | -3082 | 5 | 5 | |
| -0083 | -3083 | 5 | 6 | |
| -0084 | -3084 | 5 | 7 | |
| -0085 | -3085 | 5 | 8 | |
| -0086 | -3086 | 5 | 9 | |
| -0087 | -3087 | 5 | 10 1/ | |
| -0088 | -3088 | 5 | 10 | |

1/ Indicates continuous rotation (no stops).
2/ NS = Non shorting S = Shorting
3/ Only switches specified shall be considered qualified products.



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*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark o* ☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify):* |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

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c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBE. *Code)* – Optional |
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